

# Recovering Deleted and Damaged Files (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 03

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Time on Task:

1 hour, 53 minutes

Progress:

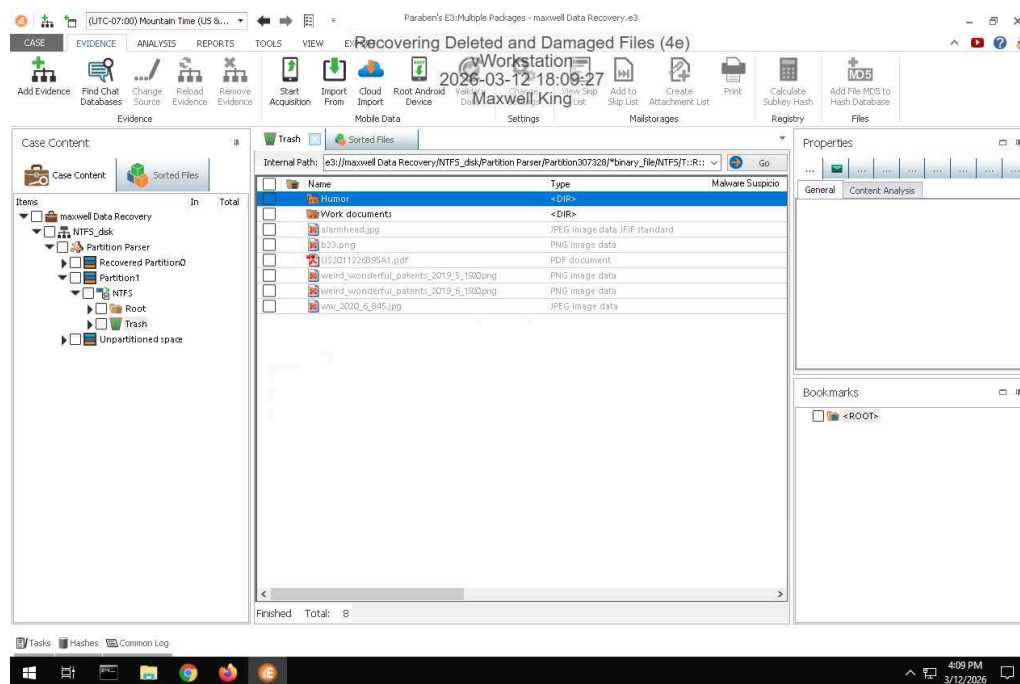
100%

Report Generated: Thursday, April 30, 2026 at 4:07 PM

## Section 1: Hands-On Demonstration

### Part 1: Recover Deleted Files from an NTFS Drive Image with E3

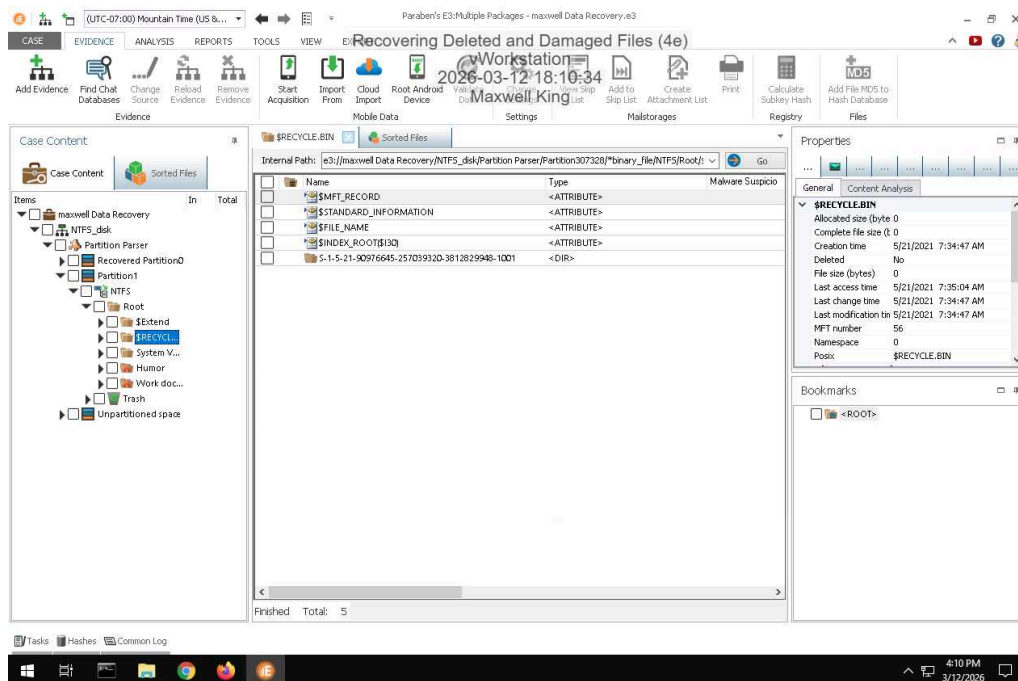
13. Make a screen capture showing the list of recovered files and folders in the E3 Trash folder.



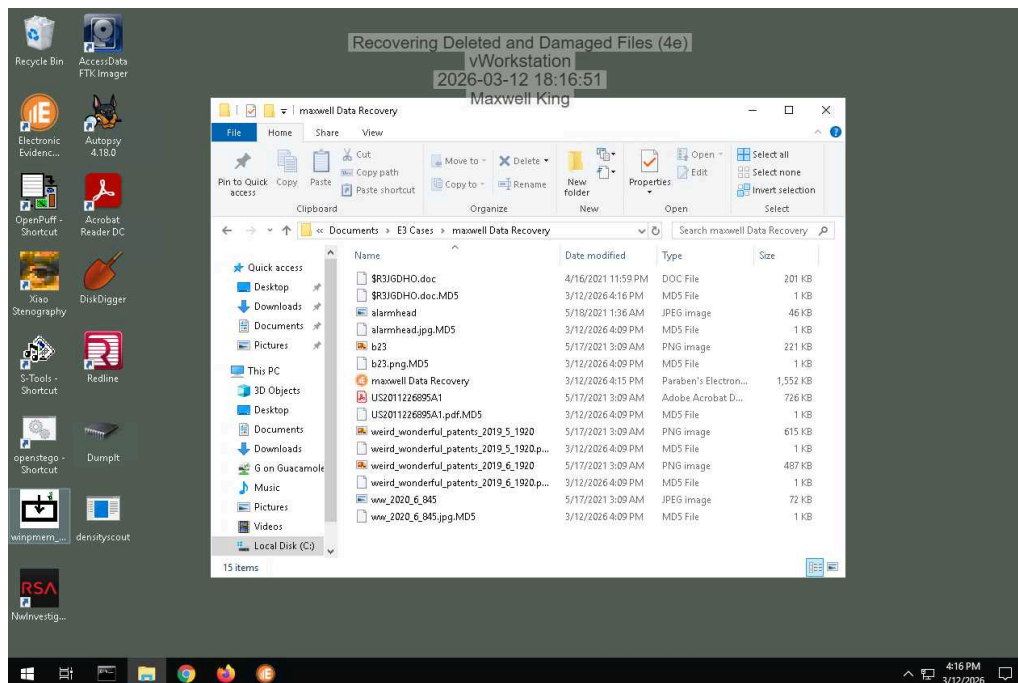
# Recovering Deleted and Damaged Files (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 03

## 20. Make a screen capture showing the patent file in the File Viewer.



## 25. Make a screen capture showing the recovered files in the File Explorer.

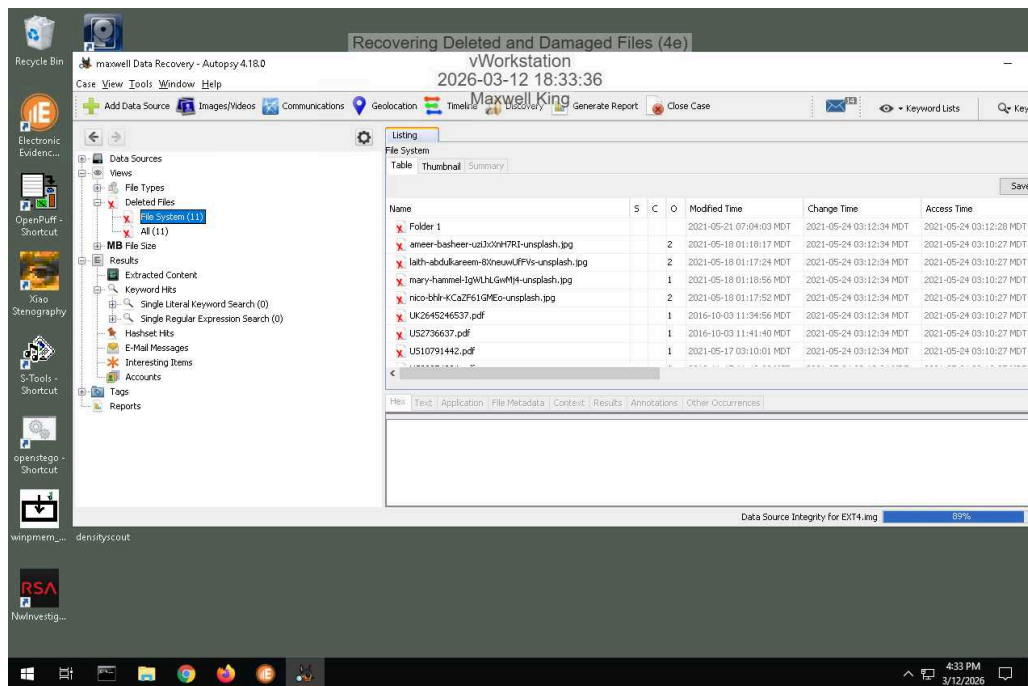


## Part 2: Recover Deleted Files from an Ext4 Drive Image with Autopsy

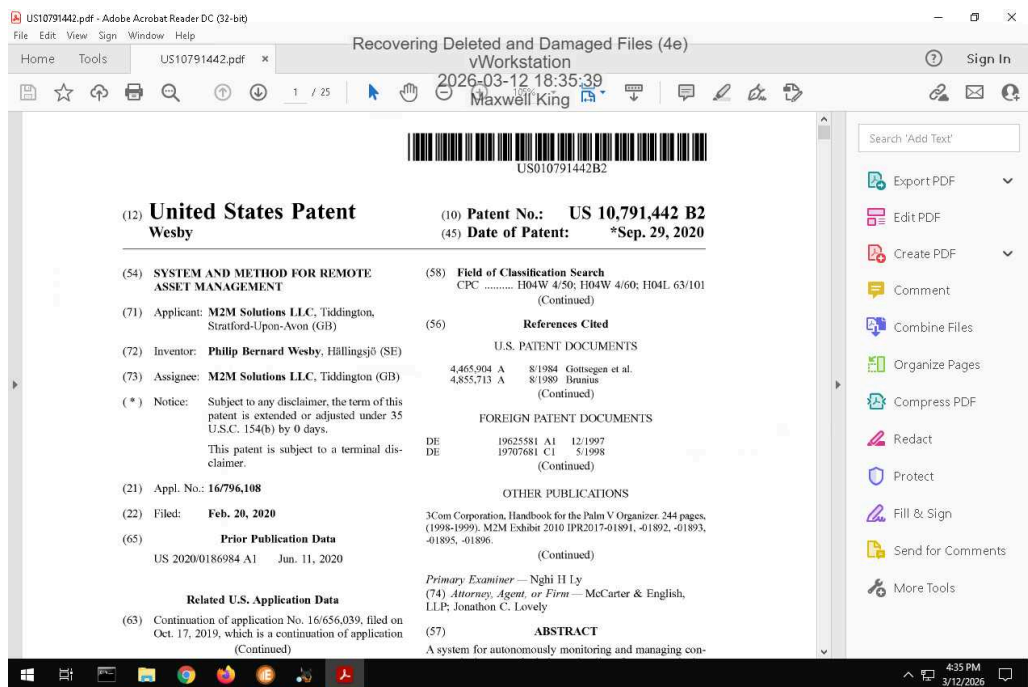
# Recovering Deleted and Damaged Files (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 03

14. Make a screen capture showing the contents of the list of deleted files in Autopsy.



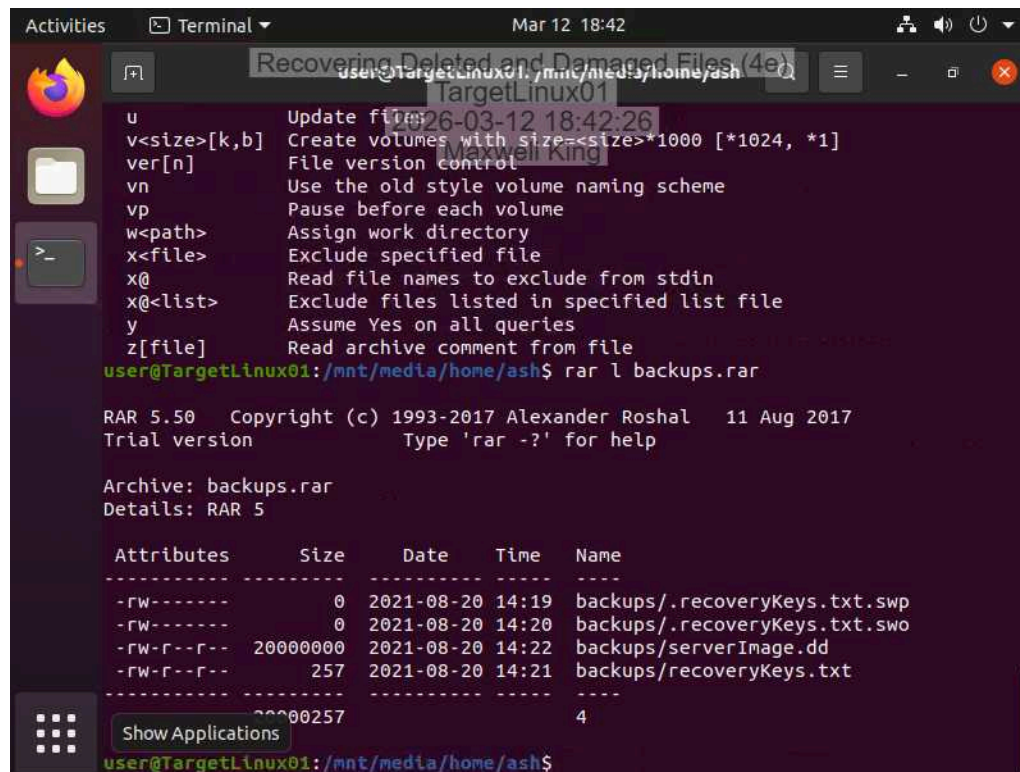
22. Make a screen capture showing the recovered patent file.



## Section 2: Applied Learning

### Part 1: Recover Deleted Files in Linux with PhotoRec

9. Make a screen capture showing the contents of the RAR archive in the `/mnt/media/home/ash` directory.



```
user@TargetLinux01:/mnt/media/home/ash$ rar l backups.rar

RAR 5.50 Copyright (c) 1993-2017 Alexander Roshal 11 Aug 2017
Trial version Type 'rar -?' for help

Archive: backups.rar
Details: RAR 5

Attributes      Size      Date      Time      Name
-----
-rw-----      0      2021-08-20 14:19 backups/.recoveryKeys.txt.swp
-rw-----      0      2021-08-20 14:20 backups/.recoveryKeys.txt.swo
-rw-r--r-- 20000000 2021-08-20 14:22 backups/serverImage.dd
-rw-r--r--   257      2021-08-20 14:21 backups/recoveryKeys.txt
-----
                20000257                4

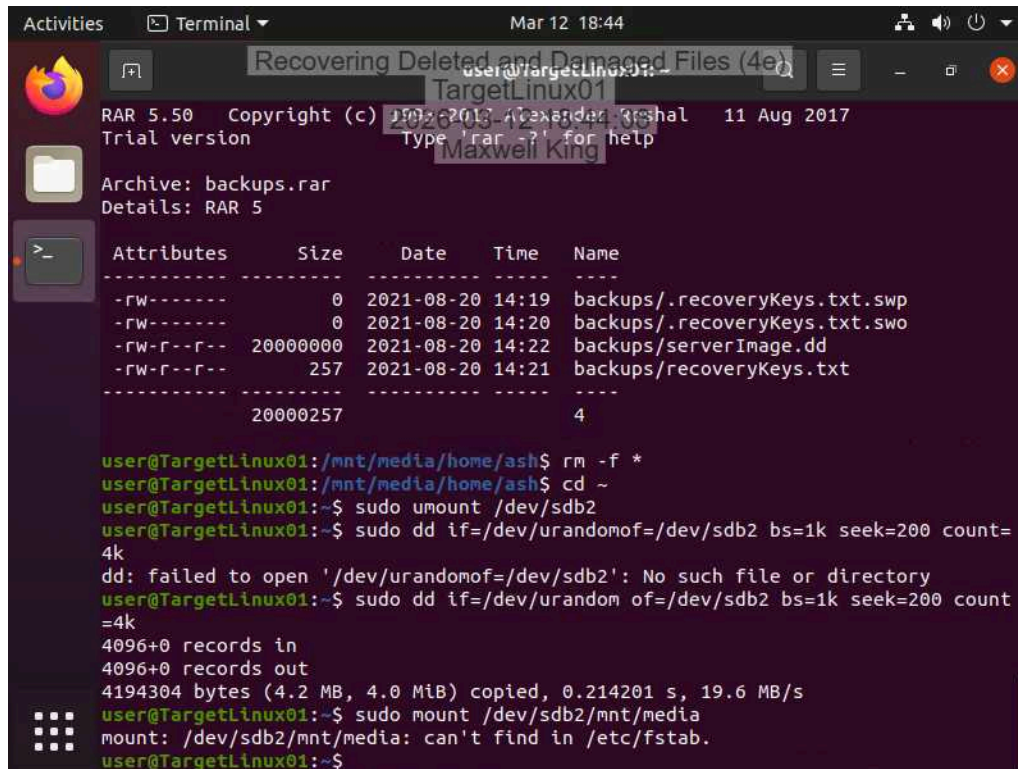
user@TargetLinux01:/mnt/media/home/ash$
```



## Recovering Deleted and Damaged Files (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 03

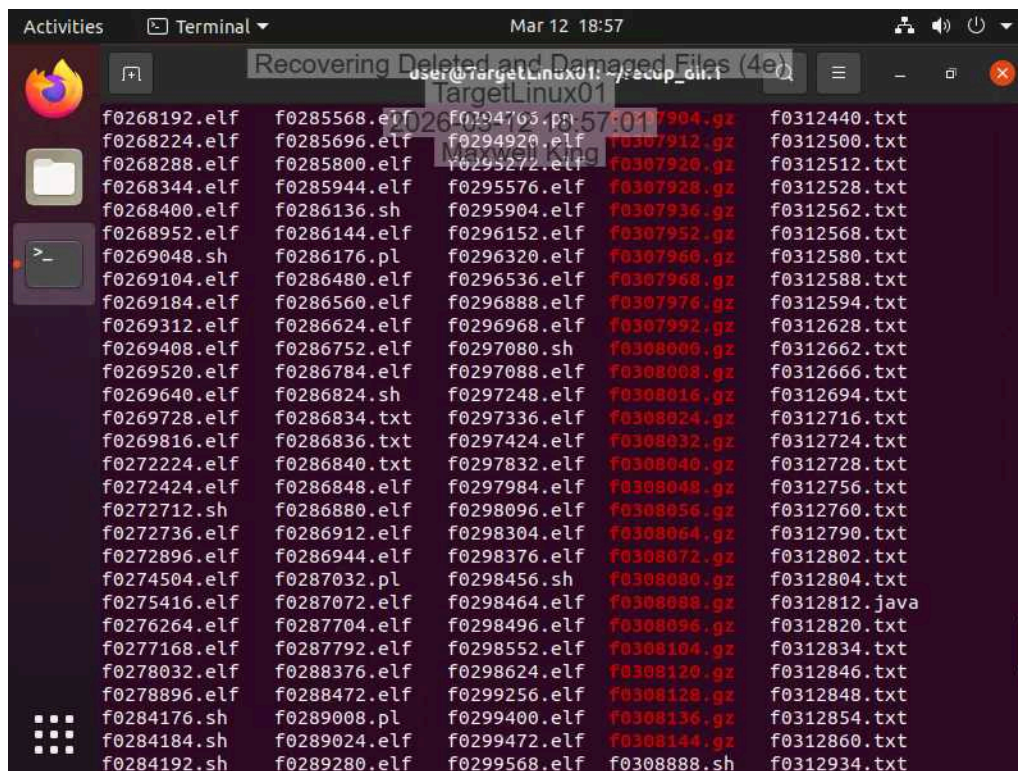
15. Make a screen capture showing the failed mount attempt on the /dev/sdb2 device.



The screenshot shows a terminal window on a system named 'TargetLinux01'. The user is in the directory '/mnt/media/home/ash'. They run 'rm -f \*' and 'cd ~'. Then they run 'sudo umount /dev/sdb2'. Next, they run 'sudo dd if=/dev/urandom of=/dev/sdb2 bs=1k seek=200 count=4k', which fails with the message 'dd: failed to open '/dev/urandom of=/dev/sdb2': No such file or directory'. They then run 'sudo dd if=/dev/urandom of=/dev/sdb2 bs=1k seek=200 count=4k', which succeeds, copying 4.2 MB. Finally, they run 'sudo mount /dev/sdb2/mnt/media', which fails with the message 'mount: /dev/sdb2/mnt/media: can't find in /etc/fstab'.

```
user@TargetLinux01:~$ rm -f *
user@TargetLinux01:~$ cd ~
user@TargetLinux01:~$ sudo umount /dev/sdb2
user@TargetLinux01:~$ sudo dd if=/dev/urandom of=/dev/sdb2 bs=1k seek=200 count=4k
dd: failed to open '/dev/urandom of=/dev/sdb2': No such file or directory
user@TargetLinux01:~$ sudo dd if=/dev/urandom of=/dev/sdb2 bs=1k seek=200 count=4k
4096+0 records in
4096+0 records out
4194304 bytes (4.2 MB, 4.0 MiB) copied, 0.214201 s, 19.6 MB/s
user@TargetLinux01:~$ sudo mount /dev/sdb2/mnt/media
mount: /dev/sdb2/mnt/media: can't find in /etc/fstab.
user@TargetLinux01:~$
```

32. Make a screen capture showing the compressed files recovered by PhotoRec.



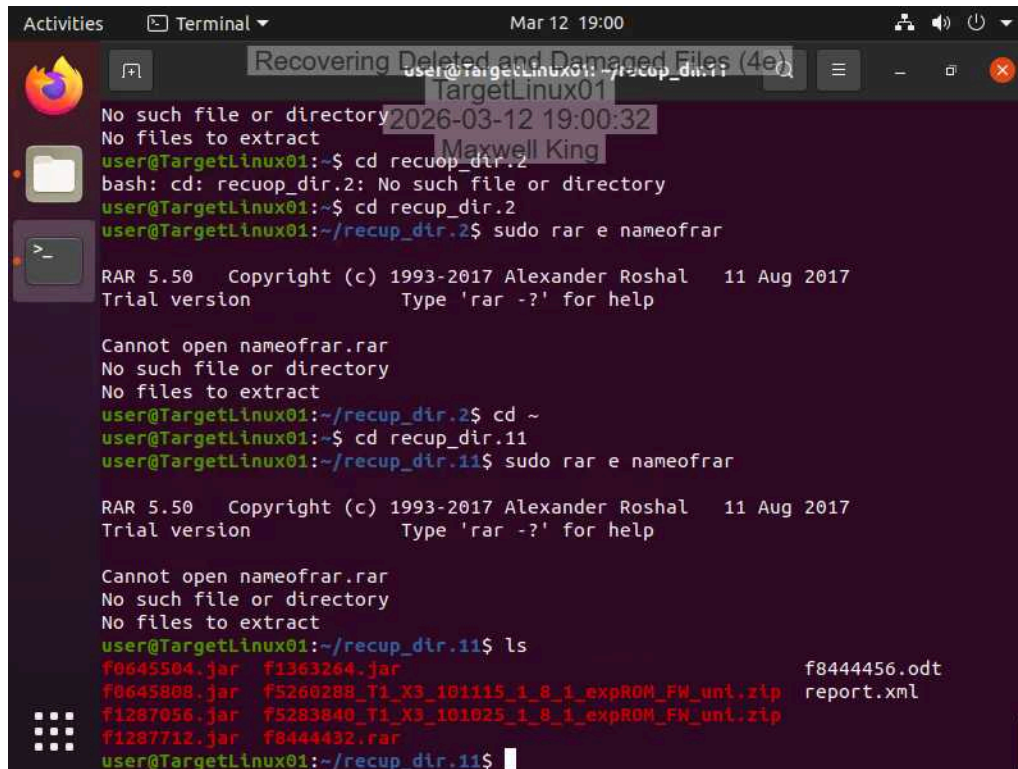
The screenshot shows a terminal window on a system named 'TargetLinux01'. The user is in the directory '/recup\_dir.1'. The terminal displays a list of recovered files, including .elf, .sh, .txt, .gz, and .java files, each with a file number (e.g., f0268192.elf, f0285568.elf, f0312440.txt, f0312500.txt, f0312512.txt, f0312528.txt, f0312562.txt, f0312568.txt, f0312580.txt, f0312588.txt, f0312594.txt, f0312628.txt, f0312662.txt, f0312666.txt, f0312694.txt, f0312716.txt, f0312724.txt, f0312728.txt, f0312756.txt, f0312760.txt, f0312790.txt, f0312802.txt, f0312804.txt, f0312812.txt, f0312820.txt, f0312834.txt, f0312846.txt, f0312848.txt, f0312854.txt, f0312860.txt, f0312934.txt).

```
user@TargetLinux01:/recup_dir.1$
f0268192.elf f0285568.elf f0294703.pl f0307904.gz f0312440.txt
f0268224.elf f0285696.elf f0294920.elf f0307912.gz f0312500.txt
f0268288.elf f0285800.elf f0295272.elf f0307920.gz f0312512.txt
f0268344.elf f0285944.elf f0295576.elf f0307928.gz f0312528.txt
f0268400.elf f0286136.sh f0295904.elf f0307936.gz f0312562.txt
f0268952.elf f0286144.elf f0296152.elf f0307952.gz f0312568.txt
f0269048.sh f0286176.pl f0296320.elf f0307960.gz f0312580.txt
f0269104.elf f0286480.elf f0296536.elf f0307968.gz f0312588.txt
f0269184.elf f0286560.elf f0296888.elf f0307976.gz f0312594.txt
f0269312.elf f0286624.elf f0296968.elf f0307992.gz f0312628.txt
f0269408.elf f0286752.elf f0297080.sh f0308000.gz f0312662.txt
f0269520.elf f0286784.elf f0297088.elf f0308008.gz f0312666.txt
f0269640.elf f0286824.sh f0297248.elf f0308016.gz f0312694.txt
f0269728.elf f0286834.txt f0297336.elf f0308024.gz f0312716.txt
f0269816.elf f0286836.txt f0297424.elf f0308032.gz f0312724.txt
f0272224.elf f0286840.txt f0297832.elf f0308040.gz f0312728.txt
f0272424.elf f0286848.elf f0297984.elf f0308048.gz f0312756.txt
f0272712.sh f0286880.elf f0298096.elf f0308056.gz f0312760.txt
f0272736.elf f0286912.elf f0298304.elf f0308064.gz f0312790.txt
f0272896.elf f0286944.elf f0298376.elf f0308072.gz f0312802.txt
f0274504.elf f0287032.pl f0298456.sh f0308080.gz f0312804.txt
f0275416.elf f0287072.elf f0298464.elf f0308088.gz f0312812.txt
f0276264.elf f0287704.elf f0298496.elf f0308096.gz f0312820.txt
f0277168.elf f0287792.elf f0298552.elf f0308104.gz f0312834.txt
f0278032.elf f0288376.elf f0298624.elf f0308120.gz f0312846.txt
f0278896.elf f0288472.elf f0299256.elf f0308128.gz f0312848.txt
f0284176.sh f0289008.pl f0299400.elf f0308136.gz f0312854.txt
f0284184.sh f0289024.elf f0299472.elf f0308144.gz f0312860.txt
f0284192.sh f0289280.elf f0299568.elf f0308888.sh f0312934.txt
```

## Recovering Deleted and Damaged Files (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 03

35. Make a screen capture showing the backup files recovered from the RAR archive.



The screenshot shows a terminal window on a Linux system. The user is attempting to extract files from a RAR archive named 'nameofrar.rar'. The terminal output shows several error messages indicating that the file or directory does not exist. The user then lists the files in the current directory, showing a list of recovered files including .jar, .zip, .odt, and .xml files.

```
Activities Terminal Mar 12 19:00
Recovering Deleted and Damaged Files (4e)
user@TargetLinux01:~/recup_dir.11$
No such file or directory
No files to extract
user@TargetLinux01:~$ cd recup_dir.2
bash: cd: recup_dir.2: No such file or directory
user@TargetLinux01:~$ cd recup_dir.2
user@TargetLinux01:~/recup_dir.2$ sudo rar e nameofrar
RAR 5.50 Copyright (c) 1993-2017 Alexander Roshal 11 Aug 2017
Trial version Type 'rar -?' for help

Cannot open nameofrar.rar
No such file or directory
No files to extract
user@TargetLinux01:~/recup_dir.2$ cd ~
user@TargetLinux01:~$ cd recup_dir.11
user@TargetLinux01:~/recup_dir.11$ sudo rar e nameofrar

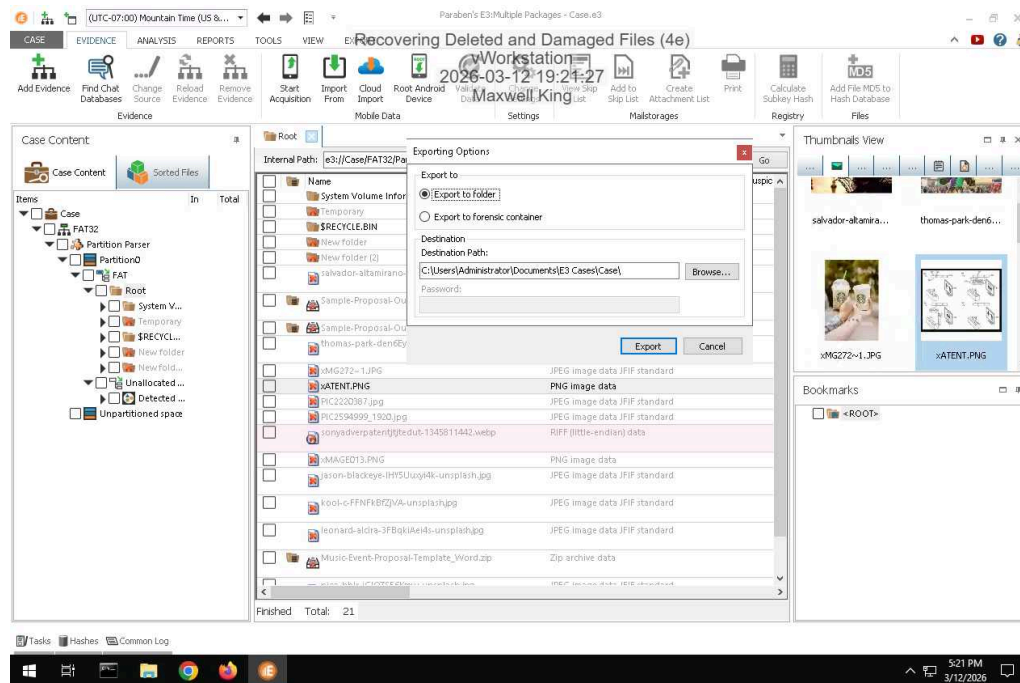
RAR 5.50 Copyright (c) 1993-2017 Alexander Roshal 11 Aug 2017
Trial version Type 'rar -?' for help

Cannot open nameofrar.rar
No such file or directory
No files to extract
user@TargetLinux01:~/recup_dir.11$ ls
f0645504.jar f1363264.jar f8444456.odt
f0645808.jar f5260288_T1_X3_101115_1_8_1_expROM_FW_unl.zip report.xml
f1287056.jar f5283840_T1_X3_101025_1_8_1_expROM_FW_unl.zip
f1287712.jar f0444432.rar
user@TargetLinux01:~/recup_dir.11$
```

### Section 3: Challenge and Analysis

#### Part 1: Recover Deleted Files from a FAT Drive Image

Make a screen capture showing the patent file recovered from the FAT32 drive image within E3.



#### Part 2: Recover Deleted Files from a APFS Drive Image

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**Make a screen capture showing the patent file recovered from the APFS drive image within Autopsy.**

